```
#include <stdio.h>

int main()
{
   int n, m;
   printf("Enter number of processes: ");
   scanf("%d",&n);

   printf("Enter number of resources: ");
   scanf("%d",&m);

   int alloc[n][m], max[n][m], need[n][m];
   int avail[m];

   printf("Enter Allocation Matrix:\n");
   for(int i=0;i<n;i++)
      for(int j=0;j<m;j++)
         scanf("%d",&alloc[i][j]);

   printf("Enter Maximum Matrix:\n");
   for(int i=0;i<n;i++)
      for(int j=0;j<m;j++)
         scanf("%d",&max[i][j]);

   printf("Enter Available Resources:\n");
   for(int i=0;i<m;i++)
      scanf("%d",&avail[i]);

   for(int i=0;i<n;i++)
      for(int j=0;j<m;j++)
         need[i][j]=max[i][j]-alloc[i][j];
```

```
int finish[n];
for(int i=0;i<n;i++)
  finish[i]=0;

int safe[n], count=0;

while(count<n)
{
  int found=0;

  for(int i=0;i<n;i++)
  {
    if(finish[i]==0)
    {
      int j;
      for(j=0;j<m;j++)
      {
        if(need[i][j]>avail[j])
          break;
      }

      if(j==m)
      {
        safe[count++]=i;

        for(int k=0;k<m;k++)
          avail[k]+=alloc[i][k];

        finish[i]=1;
        found=1;
```

```
            }
        }
    }

    if(found==0)
    {
        printf("System is not in safe state.\n");
        return 0;
    }
  }

  printf("Safe Sequence: ");
  for(int i=0;i<n;i++)
    printf("P%d ",safe[i]);

  return 0;
}
```

OUTPUT:

```
Enter number of processes: 5
Enter number of resources: 3
Enter Allocation Matrix:
0 1 0
2 0 0
3 0 2
2 1 1
0 0 2
Enter Maximum Matrix:
7 5 3
3 2 2
9 0 2
2 2 2
```

4 3 3

Enter Available Resources:

3 3 2

Safe Sequence: P1 P3 P4 P0 P2